

Trade and the Origins of the Territorial State

Johannes Boehm

November 5, 2026

What determines the emergence of territorial states?

Which economic and social forces lead to the emergence of states?
What *shapes* the geographic extent of the state?

State definition

→ Fundamental questions of political economy

Difficult to study empirically:

- Emergence of states happened a long time ago ← scarcity of data
- Geographic extent of the state is an equilibrium outcome of inter-state competition ← boundaries confound economic forces with military competition

This paper: study state formation in 10th century Eastern Europe

Why? Three reasons:

1. Formation of new territorial states out of tribal chiefdoms
   - Piast Poland, Volga Bulgaria, later Kyivan Rus
2. New territorial states are not immediately adjacent to existing territorial states
   - $\rightarrow$ realized state borders are informative about (relative) payoffs to territorial expansion, and the underlying economic and political forces
3. Novel archaeological data on commerce and on state borders of the first Polish state
   - Strongholds (dendrochronologically dated) informative about state boundaries
   - Coin finds informative about trade and other networks

This paper: study state formation in 10th century Eastern Europe

Why? Three reasons:

1. **Formation of new territorial states out of tribal chiefdoms**
   - Piast Poland, Volga Bulgaria, later Kyivan Rus
2. **New territorial states are not immediately adjacent to existing territorial states**
   - $\rightarrow$ realized state borders are informative about (relative) payoffs to territorial expansion, and the underlying economic and political forces
3. **Novel archaeological data** on commerce and on state borders of the first Polish state
   - Strongholds (dendrochronologically dated) informative about state boundaries
   - Coin finds informative about trade and other networks

Roadmap:

1. Linear regressions indicate that changes in market access and trade are related to formation of territorial states (Poland $\sim$940, Volga Bulgaria $\sim$940)
2. Model of trade and spatial extent of the state
   - Coin data to estimate geography of trade flows (cf. Boehm and Chaney, 2025)

Main findings

1. Market access contributed to state formation
   - Changing market access to Islamic silver increased the likelihood of state formation in Poland and Volga Bulgaria, around 940
2. Silver flows *shaped* the state
   - Coin flows constitute taxable traffic along rivers
   - The Piast state in Poland expanded to reach the rivers and tax silver flows
   - Borders reveal the relative returns to taxation and costs of territorial control
3. The form of revenue matters
   - River tolls: about 32% of net land revenue plus toll revenue
   - Predictable silver revenues paid soldiers, enabling further taxation, plunder, and territorial consolidation and expansion
4. Origins of Coercive Capacity among Eastern European states (Tilly '92)
   - Demand for slaves + supply of silver + geography $\Rightarrow$ spatially concentrated silver flows
   - Formation of territorial and coercive states to tax silver flows, and use coercion to get more slaves

Literature: key papers on state formation and state boundaries

- Tilly (1992), Olson (1993), Mayshar-Moav-Pascali (2022, JPE), dal Bo-Hernandez-Lagos-Mazzuca (2022, APSR), Allen-Bertazzini-Heldring (2023, AER), Heldring (2025, ARE)
- **Sanchez de la Sierra (2020, JPE)**: States as stationary bandits
  Key difference: quantification of forces from through trade data and data on border
- **Flueckiger, Larch, Ludwig, Pascali (AER forthcoming)**: Emergence of bronze age city states on trade route bottlenecks
  Key difference: here: territorial states, not city states; borders help w/ identification
- **Fernandez-Villaverde, Koyama, Lin, Sng (QJE, 2023)**: geographic determinants of state borders
  Key difference: causal evidence; equilibrium model allows counterfactuals
- **Allen (WP 2023)**: quant. model of state borders through inter-state competition
  Key difference: here: not inter-state comp. but everything else

Literature: History, Archaeology, Numismatics

Key difference: econometric identification of trade's effect on state formation; estimation of fiscal payoffs from trade flows and the state border.

- **Silver flows and the slave trade:** Noonan (1992); Jankowiak (2020)
  Changing routes and rhythms of exchange with the Islamic world
- **Silver, tribute, and Piast state formation:** Adamczyk (2014, 2020)
  Long-distance trade and silver redistribution as foundations of political power
- **Interpreting coin finds:** Brather (2007); Kilger (2008); Jankowiak (2018)
  Hoard composition and silver fragmentation as evidence of exchange
- **Territory and coercion:** Kara (2020); Biermann (2021); Izdebski et al. (2025)
  Strongholds and territorial control; slavery; social and ecological change

Political actors in Eastern Europe, 10th century



Khazar Khaganate:
6th century – 969 AD

Political actors in Eastern Europe, 10th century



Rus tribes:

Ladoga ∼ 750, Novgorod ∼ 850,

conquest of Kyiv ∼ 865;

Kyivan Rus (870s onwards)

"The Rus raid the *ṣaqāliba*, sailing in their ships until they come upon them, take them captive, and sell them in Khazaria and in Bulgar. They have no cultivated fields and they live by pillaging the land of the *ṣaqāliba*. […] They have no dwellings, villages, or cultivated fields. They earn their living by trading in sable, grey squirrel, and other furs. They sell them for silver coins which they set in belts and wear round their waists. […] They treat their slaves well and dress them suitably, because for them they are an article of trade." (Ibn Rustah, Kitāb al-Aʿlāq al-nafīsa, 903–913)

Political actors in Eastern Europe, 10th century



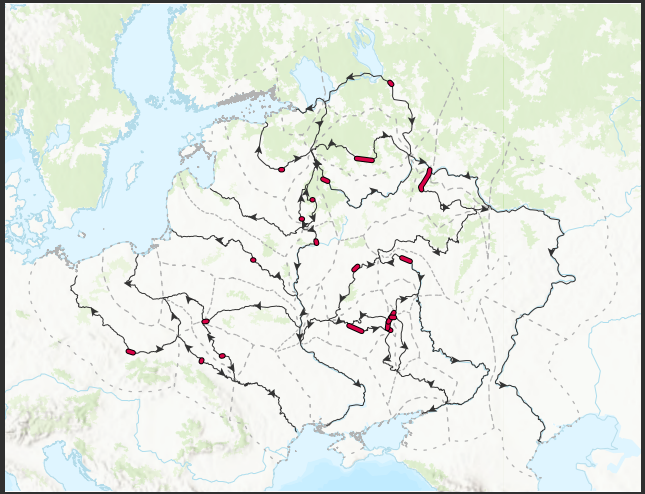
Poland (940s)

Political actors in Eastern Europe, 10th century



Volga Bulgaria (8th/9th c – 1236 AD)

Trade routes in Eastern Europe, 9th–10th century



Slaves, furs, and forest products exchanged for Arab silver

- Overland routes in red
- Rivers in black; arrows show flow direction
- Dashed lines partition space by nearest trade route (up to 200 km)

Sources: Леонтьев and Носов, 2012; Tentiuc, 2020; Channon and Hudson, 1995.

Around 940 AD: Formation of Polish state under Piast dynasty (Mieszko I.)

Ibn Yaqub writes, around 965:

> *"Regarding the land of Mieszko, this is the largest of the Slavs' lands. It is abundant with food – meat, honey and crops. He collects taxes in the form of [al–mathaqil al–marqktija] (silver, by weight). and they are used to pay the monthly salaries of his men [warriors], each of whom receives a fixed number. He has three thousand armoured men and they are divided into military units. One hundred of them is equal to a thousand other [warriors]. He gives them clothing, horses, weapons and everything else they need." [transl., after Kowalski, 2022]*

Hence:

1. Taxation of trade flows on rivers (main taxable source of silver; Adamczyk 2014, Hardt, 2018)
2. Silver used to pay warriors
3. Archaeological evidence shows that Poland was heavily and actively involved in the slave trade as well (Adamczyk, 2014, Roach, 2020, Biermann, 2021)

The setting: Eastern Europe in the 8th-10th century

Extremely data-scarce setting:

1. Literary sources:
   - Chronicles: Frankish monks (Bavarian Geographer, ca. 830-870), Byzantine chronicles (Constantine VII's *De Administrando Imperio* ∼950)
   - Arab travel reports (Ibn Yaqub ∼960s, Ibn Fadlan 922), Khazar letters (King Joseph's letter ∼950s)
   - Later chronicles: Rus *Primary Chronicle*: 12th century, Gallus Anonymus: 12th century.
   - Only "official" document is *Dagome Iudex* letter (∼991)

2. Archaeological sources:
   - Coins
   - Metal tools, weapons, jewellery (settlements), objects of exchange (glass beads)
   - Pottery
   - Strongholds (dated w. tree-rings)
   - Palnyology (agriculture/deforestation)

Data

$1{,}255$ coin hoards with $193{,}711$ coins, digitized from the numismatic & archaeological literature

- Hoard *tpq* between 725 and 1040 AD
- From present-day Poland, Ukraine, and Moldova (west) to the Volga basin (east)

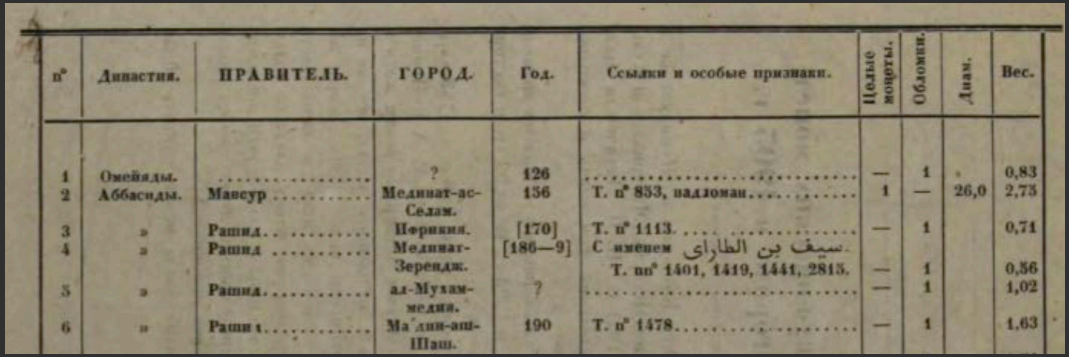
Coins & hoards show:

- *Where* and *when* a coin was minted
- *Where* and *when* it was buried in the form of a hoard (*terminus post quem*)

Structural econometrics (later) add hoards from Gotland and Bornholm.

▸ Map ▸ Coin statistics ▸ Hoard statistics

Example of a hoard description



№	Династия.	ПРАВИТЕЛЬ.	ГОРОД.	Год.	Ссылки и особые признаки.	Целые монеты.	Обломки.	Диам.	Вес.
1	Омейяды.	.....	?	126	.....	—	1	.	0,83
2	Аббасиды.	Мансур	Мединат-ас-Селам.	156	Т. № 853, надлом.....	1	—	26,0	2,73
3	»	Рашид..	Ифрикия.	[170]	Т. № 1113.	—	1		0,71
4	»	Рашид	Мединат-Зереджк.	[186—9]	С именем سيف بن الطارای. Т. №№ 1401, 1419, 1441, 2815.	—	1		0,56
5	»	Рашид.....	ал-Мухаммедия.	?	.....	—	1		1,02
6	»	Рашид.....	Ма'дин-аш-Шаш.	190	Т. № 1478.....	—	1		1,63

Bykov (1925): a hoard found in Novgorod in 1903. Columns report dynasty, ruler, mint, mint year (*anno hegirae*), notes, whole coins, fragments, diameter, and weight.

All coins are Islamic dirhams; hoard *tpq*: 341 AH (952–953 AD).

What do coin hoards tell us?

1. **Presence and quantity of finds**: coarse measures of access to silver
   - But q'ty also depends on deposition, discovery, and documentation
2. **Composition by origin and age**: more informative about circulation
   - Coins valued by silver weight → fungible across origins and vintages
   - More young coins → larger recent inflows relative to the existing stock

Long-distance coin flows are largely the result of commerce (Noonan, 1992); Short-distance flows of coins can be due to other reasons as well (plunder, gift, tribute, ransom,...).

I'll use the term "trade" in a broad sense, encompassing these other motives.

▸ Temporal distributions

Reduced-form: relationship between market access/trade and state formation?

Challenge: Trade and state formation may be jointly determined

- Productivity could drive both
- States may facilitate trade $\rightarrow$ reverse causality

Strategy:

1. OLS regressions of state presence on trade indicators from coins
2. IV regressions using shifts in market access:
    - Rus–Khazar conflict: disrupts established routes through Khazaria
    - Silver supply shifts: production shifts from the Middle East to Central Asia

Regression specifications

Region $\times$ 25-year period:

$$\text{StatePresent}_{it} = \beta\,\text{Coins}_{it} + \alpha_i + \alpha_t + \varepsilon_{it}$$

Hoard level:

$$\text{StatePresent}_h = \gamma\,\text{YoungShare}_h + \psi(\text{Size}_h) + \alpha_{i(h)} + \alpha_{t(h)} + \varepsilon_h$$

Coin level:

$$\text{StatePresent}_{h(c)} = \delta \log(\text{Age at deposit})_c + \alpha_{i(h(c))} + \alpha_{t(h(c))} + \varepsilon_c$$

YoungShare: fraction younger than 20 or 40 years; ψ: cubic in log hoard size.

StatePresent: Location contains a territorial state (Poland or Volga Bulgaria after 940)

Region: Area of closest proximity to a given river segment (up to 200km)

OLS: correlation between state formation and coin inflows

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_h$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0181**				
	(0.0033)				
log Number of Coins		0.0166$^+$			
		(0.0087)			
Fraction of coins $<$ 20 y.o.			0.120**		
			(0.027)		
Fraction of coins $<$ 40 y.o.				0.141**	
				(0.028)	
Log coin age at deposit					-0.0191**
					(0.0026)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
R^2	0.599	0.398	0.820	0.821	0.908
Observations	429	429	911	911	105597

$^+$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

SEs clustered by region in (1)–(2), by coin group in (5).

More coins and younger coins are associated with state presence.

Instrumental variables: IV #1

Rus–Khazar conflict: disruption of access to Arab traders

$$Z_{it}^{\text{conflict}} = \log\left(\min_{a \in \{\text{Black Sea, Caspian Sea, Bolgar}\}} \text{TravelTime}_{ia,t}\right)$$

- Conflict dated to 925 in the empirical implementation
- From 925: travel speeds through Khazar territory are *halved*
- Baseline speeds: 8 km/h downstream; 1 km/h upstream and at portages

  ▸ Experimental archaeology
- Regions relying on disrupted routes lose relative market access

Identifying assumption: conditional on fixed effects, this exposure potentially affects state formation *only* through market access.

IV results: trade disruption

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_h$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0324**				
	(0.011)				
log Number of Coins		0.100*			
		(0.042)			
Fraction of coins $<$ 20 y.o.			2.751**		
			(1.06)		
Fraction of coins $<$ 40 y.o.				2.682**	
				(0.97)	
Log coin age at deposit					-0.420**
					(0.094)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.937	6.085	6.802	7.703	17.50
R^2	0.142	-1.738	-10.49	-9.088	-9.147
Observations	429	429	911	911	105597

$^+$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Trade indicator instrumented by log shortest travel time to Arab traders. SEs clustered by region in (1)–(2), by coin group in (5).

First stages are weak in some specifications.

Instrumental variables: IV #2

Silver supply shifts: Abbasid to Samanid coin production

- From the late 9th century, silver inflows shift toward Central Asia
- Regions differ in proximity to the changing sources of coin supply

$$\text{SilverMarketAccess}_{it} = \sum_{j} \frac{d_{ij}^{-1}}{\sum_{j'} d_{ij'}^{-1}} s_{jt}$$

- d_{ij}: great-circle distance to the centroid of polity j's mints
- s_{jt}: polity j's share of coins minted in period t and found in the data

Identifying assumption: external silver-supply shifts potentially affect state formation *only* through market access.

▸ Mint maps

IV results: silver market access

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_h$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0173**				
	(0.0016)				
log Number of Coins		0.0762*			
		(0.028)			
Fraction of coins < 20 y.o.			3.963**		
			(1.42)		
Fraction of coins < 40 y.o.				3.942**	
				(0.94)	
Log coin age at deposit					-0.476**
					(0.11)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Unit of Observation	Region × Time	Region × Time	Hoard	Hoard	Coin
First stage w. id. robust F	4.044	7.376	12.05	13.11	55.04
R^2	0.380	-0.845	-22.41	-20.39	-11.85
Observations	429	429	911	911	105597

$^+$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Exposure-robust SEs from equivalent shock-level regressions (Borusyak, Hull, and Jaravel, 2022).

Positive effects for coin inflows; negative effect for coin age.

Interpreting the reduced-form evidence

- Both instruments support a causal role for access to silver
- Could state formation have triggered the conflict?
  - Historical accounts emphasize Byzantine involvement, not Piast or Bulgar expansion
- Could conflict directly increase demand for a state?
  - Poland remote from the conflict; harder to rule out for Volga Bulgaria
  - Silver-supply IV provides a second source of variation

None of this says that there is only *one* determinant of territorial state formation.

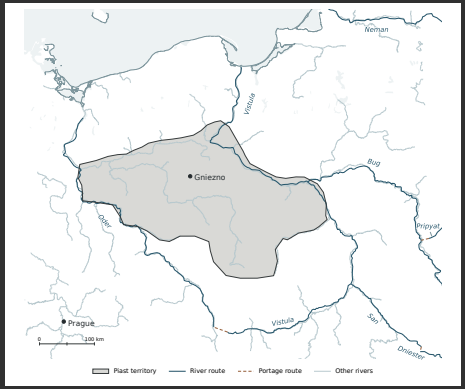
▸ Exposure controls ▸ Including Rus territories

Part II: Silver, market access, and the shape of the early Piast state in Poland

To better understand the mechanisms that led to the rise of territorial states, we'll focus on one case:

The emergence of the first Polish state under the Piast dynasty and Duke Mieszko I.

First Polish state under the Piast dynasty



Poland, ca. 960. Source: Kara (2020).

Original

1. Territory reaches the Vistula and Oder
   Taxable trade; defensible frontiers
2. Limited expansion toward the Baltic Sea
   Exposure to Viking raids
3. Gniezno lies between the main river corridors
   Cost of collecting revenue at a distance
4. Uneven expansion in the south
   Potential competition with Bohemia

Model components

1. Estimate geography of inter-regional coin flows
   - How large was silver “traffic” along the rivers?
   - Coins of different origins and vintages trace flows
   - Flows can be interpreted as the result of trade (Boehm and Chaney, 2025), plunder, or other motives
   - Estimate coin stocks and “trade” shares from hoard counts and composition
   - Assign flows to shortest paths $\rightarrow$ tax base along each river

2. Model of territorial choice within Poland
   - Given tax base along rivers, ruler chooses territory to maximize revenues net of costs
   - Land taxation and river tolls, collection costs, political competition, frontier defense
   - Fit the observed border $\rightarrow$ relative importance of these forces

Inter-regional model: household budget constraint

Running out of time...

Households in region i enter period t with a stock of coins $S_i(t)$:

$$S_i(t+1) = \underbrace{(1-\lambda)S_i(t)}_{\text{Surviving coins}} + \underbrace{M_i(t)}_{\text{New minting}} + \underbrace{Y_i(t)}_{\text{Income}} - \underbrace{X_i(t)}_{\text{Expenditure}}$$

- λ: fraction of inherited coins lost, melted, or converted into objects
- $M_i(t) \geq 0$: exogenous *mint mass*, i.e. new coins issued in i at t; zero outside minting regions
- Consumption expenditure $X_i(t)$ needs to be financed with coins ($\rightarrow$ outflow of coins)
- Income $Y_i(t) = w_i(t)L_i(t)$ received at the end of the period ($\rightarrow$ inflow of coins)

Assume no saving out of available balances, $s_i(t) = 0$: then

$$X_i(t) = (1-\lambda)S_i(t) + M_i(t) \quad \Rightarrow \quad S_i(t+1) = Y_i(t)$$

Accounting for coin types

A coin type is a *mint location* $\times$ *mint period*

$S_{mi}(\tau, t)$: inherited coins in i at t, minted in m at $\tau < t$.

Fungibility: agents blind to vintages $(m, \tau) \Rightarrow$ each coin type is "spent" in the same proportions. The *flow* of coins of each type is proportional to the *stock*.

$$S_{mi}(\tau, t+1) = \sum_n (1-\lambda) S_{mn}(\tau, t) \pi_{ni}(t), \qquad \tau < t.$$

New types enter at their mint and mint date: $S_{mm}(t, t) = M_m(t)$. Here

$$\pi_{ni}(t) = \frac{X_{ni}(t)}{X_n(t)} : \quad n\text{'s expenditure share on goods from } i\text{; coins move } n \to i.$$

Data-generating process

1. Hoard counts discipline the scale of coin stocks

$$D_i(t) \sim \text{Poisson}(\psi S_i(t))$$

- Assumption: common formation/discovery rate ψ across regions and periods

2. Hoard composition disciplines coin-type shares

$$q_h \sim \text{Dirichlet}(\kappa H_i(t)), \qquad S_h \mid q_h \sim \text{Multinomial}(N_h, q_h)$$

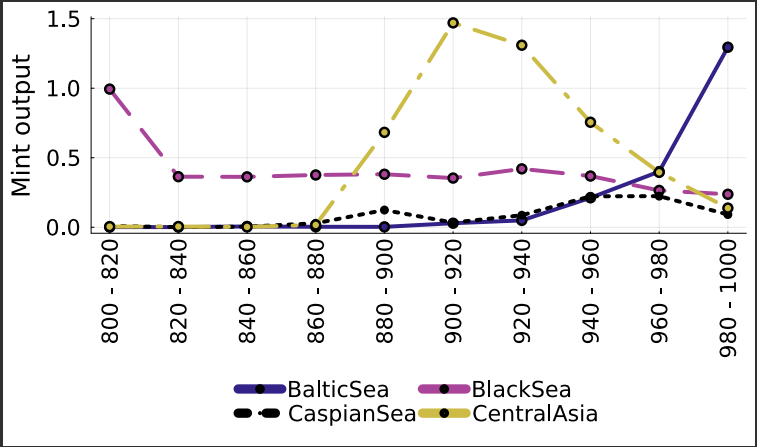
- $H_i(t)$: vector of population type shares $S_{mi}(\tau, t)/S_i(t)$; N_h: number of coins in hoard h
- Finite κ allows coins within a hoard to be more similar than independent draws

Estimate jointly by maximum likelihood. $\hat{\kappa} = 4.99$. ▸ Evidence

Implementation of the trade model

1. Geography and time
   - 13 interior regions and 4 peripheral minting regions
   - Baltic Sea, Central Asia, Black Sea, Caspian Sea
   - 20-year periods; coin types defined by mint region and mint period
2. Estimation
   - Period-specific mint masses and trade shares; overdispersion parameter
   - Nonzero trade: interior/Baltic goods exporters selling to minting regions
   - Hoards from 1000–1039 help identify earlier flows; report estimates before 1000
3. River traffic
   - Recover expenditures and trade flows from coin stocks and shares
   - Assign each flow to its shortest path through the river network

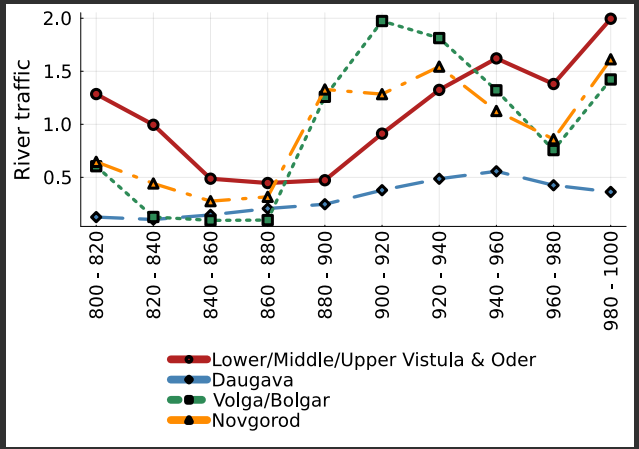
Estimated sources of silver: Mint volumes $M_i(t)$



Central Asian inflows rise; Baltic inflows become sizable late in the 10th century.

Estimated mint masses reaching the study area, not total production at the mints.

Estimated river traffic



Polish river traffic rises from 900–920 and exceeds the other plotted regions from 940.

▸ Polish estimates ▸ Alternative specification

Territorial choice block: revenues

Poland is a graph of locations V and directed edges E. Choose which cells to claim, $x_i \in \{0, 1\}$; Gniezno (c) is always claimed, $x_c = 1$.

Ruler's problem: max revenue minus costs $\max_{x \in \{0,1\}^{|V|} : x_c = 1} \Pi(x)$ (Alesina-Spolaore '97).

Production tax revenue, net of collection costs
A claimed cell yields 1 before transport; value falls with return travel time $\mathfrak{t}_{ci}$ to the capital, at rate σ.

$$+ \sum_{i \in V} (1 - \sigma \mathfrak{t}_{ci}) x_i$$

River toll revenue
Claiming a river cell yields $\tau \Xi_{r(i)}$: toll payoff per unit of traffic τ times estimated traffic $\Xi_{r(i)}$.

$$+ \sum_{i \in V} \tau \Xi_{r(i)} x_i$$

River traffic is taken as given; $\Xi_{r(i)} = 0$ away from taxable rivers.

Territorial choice block: costs

Viking raids
Expected loss on claimed land: ω at the Baltic coast, declining with travel time t_i^{sea} from the sea.

$$-\sum_{i \in V} \omega e^{-t_i^{\text{sea}}/\kappa} x_i$$

Competition with Bohemia
Cost is ρ at Prague and declines with return travel time t_i^{prague}; may reflect a behavioral wedge rather than a resource cost.

$$-\sum_{i \in V} \rho e^{-t_i^{\text{prague}}/\kappa} x_i$$

Frontier defense
Cost λ per edge from claimed to unclaimed land; discounted by $e^{-\beta_R}$ on eligible sides of held river cells.

$$-\sum_{(i,j) \in E} \lambda e^{-\beta_R R_{ij}} x_i (1 - x_j)$$

Estimation and computation

Choose parameters to minimize disagreement with the digitized Piast territory $\bar{x}$:

$$\hat{\theta} \in \arg\min_{\theta} \sum_{i} \mathbf{1}\{\bar{x}_i \neq x_i^*(\theta)\}$$

- Equivalent to MLE under independent cell misclassification with a common probability below $1/2$
- 73-by-62 grid; 10 km cells; Gniezno fixed as capital
  Combinatorial optimization problem: 2^{4525} choices, but:
- Costly frontier defense ($\lambda > 0$) makes the ruler's problem *graph-cut representable* and solvable by a minimum cut in almost-linear time

▸ Min-cut = max-flow

What identifies the model parameters?

1. **Overall extent from Gniezno** $\rightarrow$ spatial decay σ
2. **Reaching rivers with different traffic volumes** $\rightarrow$ toll payoff τ
3. **Northern reach toward the Baltic** $\rightarrow$ Viking cost ω, decay κ
4. **Southern reach toward Bohemia** $\rightarrow$ competition cost ρ, decay κ
5. **Compactness and alignment with rivers** $\rightarrow$ frontier cost λ, river discount β_R

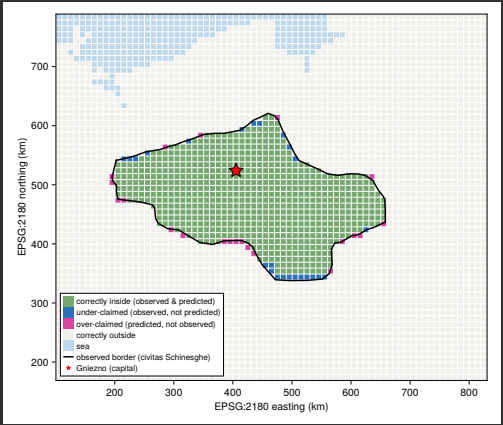
The border data recovers *relative* payoffs in the objective function.

▸ Counterfactual borders

Parameter estimates and model fit

1. **Production taxes**: collection breaks even at 14 travel days from Gniezno
2. **River tolls**: a high-traffic river cell is worth 7.1 ordinary land cells
3. **Viking threat**: coastal cost is 3.8 cell tax revenues; halves every 2.3 travel days
4. **Frontier defense**: 1.9 cell tax revenues per land edge; 39% of this on eligible sides of held river cells

▸ Full parameter table



The composition of the ruler's payoff

Description	Mathematical object	Value	% of ($\star$)	resource or shadow cost?	in-kind or monetary?
Net revenues from land tax ($\star$)	$\sum_{i \in V}(1-\sigma \mathbf{t}_{ci})x_i$	315.4	100.0	resource	revenues in-kind costs monetary
Revenues from river tolls	$\sum_{i \in V} \tau_r \Xi_{r(i)} x_i$	145.7	46.2	resource	monetary
Foreign political competition cost	$-\sum_{i \in V}\left(\omega e^{-\mathrm{t}_i^{\mathrm{sea}}/\kappa} + \rho e^{-\mathrm{t}_i^{\mathrm{prague}}/\kappa}\right) x_i$	-169.7	-53.8	either possible	either possible
Frontier defense	$-\sum_{(i,j)\in E} \lambda e^{-\beta_R R_{ij}} x_i (1-x_j)$	-170.8	-54.2	resource	monetary
Sum	Π	120.60	38.2		

- Land revenues are the largest component, predominantly in kind
- River tolls are monetary: 146 units, or 32% of land plus toll revenues
- Political competition may capture behavioral wedges as well as resource costs

Linking river tolls to macro estimates of silver inflows

The border identifies *relative* fiscal returns, not the level of silver receipts.

Additional assumption: on average, 10% of river traffic is paid as a silver toll

- Documented for Khazars and Volga Bulgars (although not known for Poland)

Combining the trade and territorial models, averaged over 940, 960, and 980:

- River tolls: about 0.16 units; total coin inflows: about 0.7 units

→ Roughly one quarter of silver from tolls, three quarters from other sources (most likely exports of slaves and forest products)

Origins of the territorial state: what did we learn?

1. Market access change made river locations valuable and triggered state formation
   - Route disruptions and shifts in Islamic coin production redirected silver
   - Market access change increased return to controlling rivers
2. Market access mattered for both emergence and territorial extent
   - Regional evidence: improved silver access increased state formation
   - Territorial model: taxable traffic helped draw the Piasts toward the rivers
3. Access to money/silver was not enough for territorial state formation
   - Example of Rus (who had silver but developed a territorial state only much later) shows that money/silver was not enough
   - Most likely explanation is that silver from tribute/plunder was not sufficiently predictable to allow switch to paid retainers
4. A small silver revenue stream could increase coercion & a larger tax base
   - Land taxes were largely in kind; the military retinue was paid in silver
   - Paid warriors collected taxes, defended border, obtained captives and export goods
   - Long-term impact of market access shocks

Trade financed coercion $\rightarrow$ coercion generated further tax and export revenues.

Origins of Coercive Capacity among Eastern European States (Tilly '92)

Tilly (1992): Eastern European states were first coercion-intensive; Western European states were first "capital"-intensive, and build coercive capacity (military) by taxing "capital".

Tilly (1992) takes existence of states as given (990 CE – 1992 CE).

This paper shows that the coercive nature of Eastern European states is intimately linked to their formation:

- Demand for slaves + availability of silver, concentrated river geography
- Emergence of coercive territorial states to tax silver flows
- Coercion was also used to obtain more slaves (and sell them for silver)

Still, commerce was what initially gave rise to coercion.

THANK YOU!

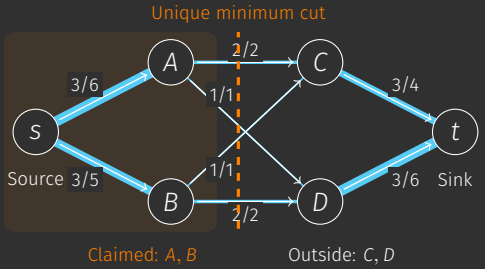
johannes.boehm@unige.ch

BACKUP SLIDES

Territorial choice as a min-cut problem

Let v_i be land and toll revenue net of collection and threat costs, and $d_{ij} = \lambda e^{-\beta_R R_{ij}}$ the cost of a frontier edge:

$$\Pi(x) = \sum_i v_i x_i - \sum_{(i,j)\in E} d_{ij} x_i (1 - x_j).$$



Width = capacity; label = flow/capacity.

Crossing pipes do not meet. Illustrative values.

Encode the ruler's payoffs

$v_i > 0$: pipe $s \to i$ of capacity v_i.
$v_i < 0$: pipe $i \to t$ of capacity $-v_i$.
$i \to j$: capacity d_{ij} for frontier defense.

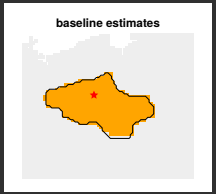
The cut is the border.

Source = Capital;
Sink = a point far outside

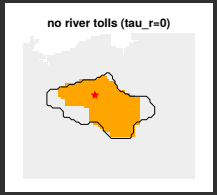
Back

Counterfactual borders

Baseline



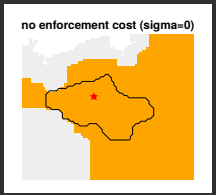
No river tolls ($\tau = 0$)



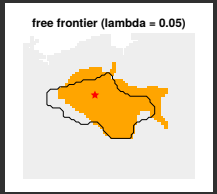
No Viking threat ($\omega = 0$)



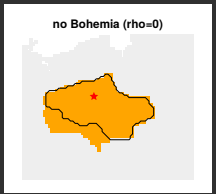
No spatial decay ($\sigma = 0$)



Low defense cost ($\lambda = 0.05$)



No Bohemia ($\rho = 0$)

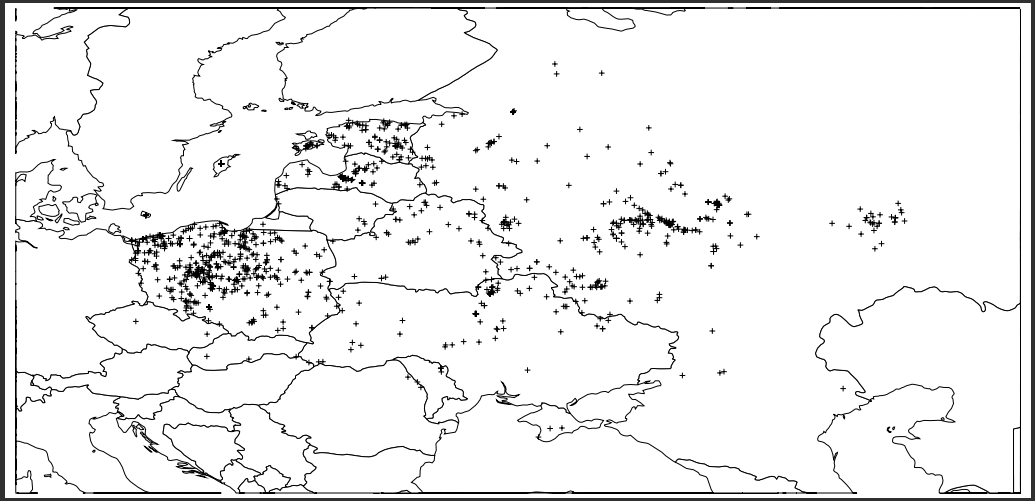


Orange: predicted territory; black: observed border; star: Gniezno.

Other parameters and river traffic held fixed.

Back

Geographic distribution of coin hoards



Back

Coin summary statistics

	count	mean	sd	min	p10	p50	p90	max
Has mint date interval	152,715	0.59	0.49	0	0	1	1	1
Has mint location	152,715	0.44	0.50	0	0	0	1	1
Has mint location and date interval	152,715	0.42	0.49	0	0	0	1	1
Mint date interval, years	90,060	17.96	35.44	0	0	1	60	200
Mint date interval, start year	90,423	883.88	72.64	54	775	904	967	1,025
Mint date interval, end year	90,415	901.73	68.88	57	799	913	982	1,200
Age at tpq	90,060	37.65	38.98	-100	4	30	82	929
Has material	152,715	0.61	0.49	0	0	1	1	1
Coin is gold	152,715	0.00	0.01	0	0	0	0	1
Coin is silver	152,715	0.61	0.49	0	0	1	1	1
Coin is copper/bronze	152,715	0.00	0.02	0	0	0	0	1
Has denomination	152,715	0.88	0.33	0	0	1	1	1
Has some empire/dynasty information	152,715	0.99	0.07	0	1	1	1	1
Geodesic distance mint to hoard, km	67,505	2,863.50	1,002.24	25	833	3,058	3,982	5,051

Tabulated sample: 2 coins. Smaller sample than the full dataset.

Back

Hoard summary statistics

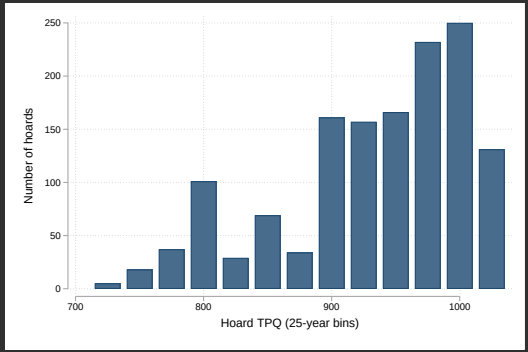
	count	mean	sd	min	p10	p50	p90	max
Hoard tpq	1,067	921.27	66.44	743	811	939	996	1,000
Number of coins in hoard	1,067	143.13	762.94	1	1	8	284	20,966
Fraction of coins with mint date interval	1,067	0.84	0.26	0	0	1	1	1
Fraction of coins with mint location	1,067	0.55	0.40	0	0	1	1	1
Fraction of coins with mint date interval and mint location	1,067	0.53	0.40	0	0	1	1	1
Average mint date interval	1,066	16.44	30.16	0	1	4	50	200
Average age of coins at tpq	1,066	21.98	32.79	-100	-0	16	58	464
Fraction of coins with material	1,067	0.70	0.46	0	0	1	1	1
Fraction of coins that are gold	1,067	0.00	0.04	0	0	0	0	1
Fraction of coins that are silver	1,067	0.68	0.46	0	0	1	1	1
Fraction of coins that are bronze	1,067	0.01	0.11	0	0	0	0	1
Fraction of coins with denomination	1,067	0.85	0.34	0	0	1	1	1
Fraction of coins with empire/dynasty information	1,067	1.00	0.01	1	1	1	1	1
Average distance of coins from mint, km	854	2,806.74	960.87	286	1,022	2,980	3,910	4,443

Tabulated sample: 2 hoards, *tpq* 5–9. The full dataset covers 725–1040.

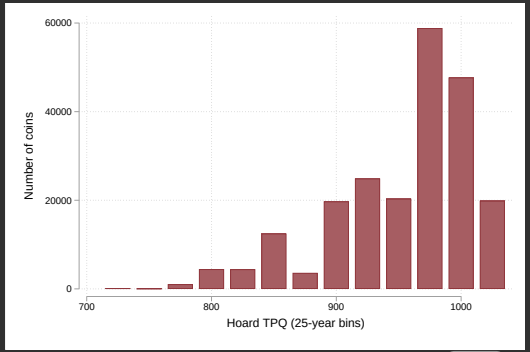
Back

Temporal distribution of hoards and coins

Hoards by *tpq*

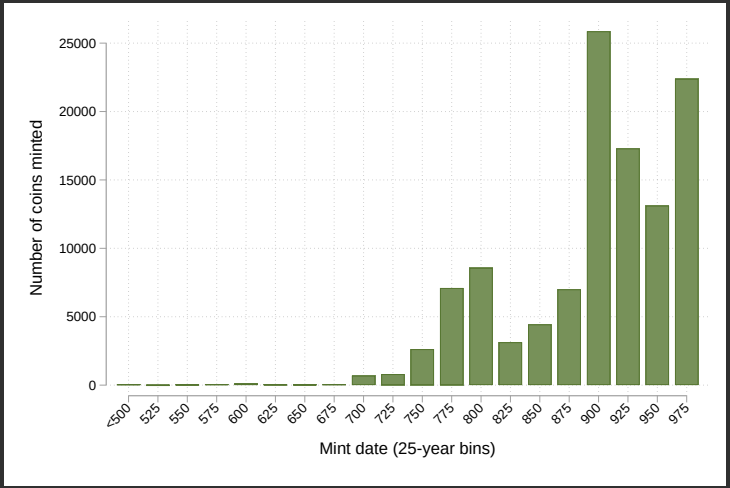


Coins by hoard *tpq*



Back

Distribution of coin mint dates

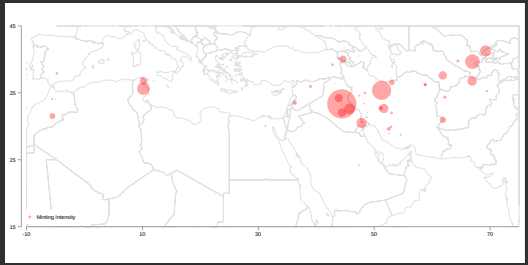


Interval-dated coins use the midpoint of the mint-date interval.

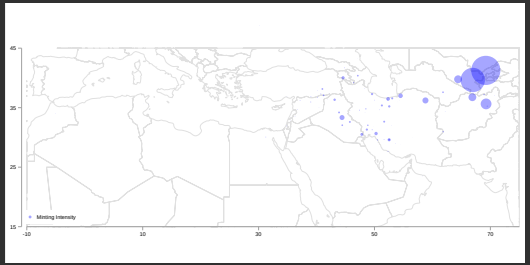
Back

Origins of Islamic silver: before and after 875

Before 875



After 875

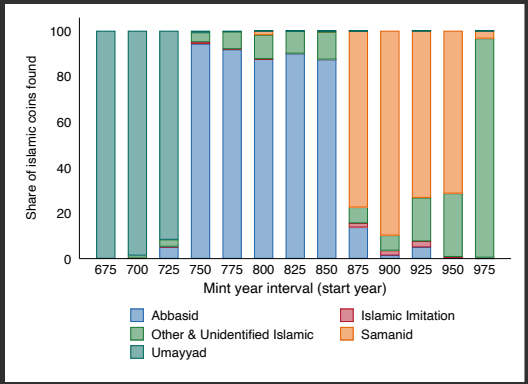


Coins with known mint locations and mint years.

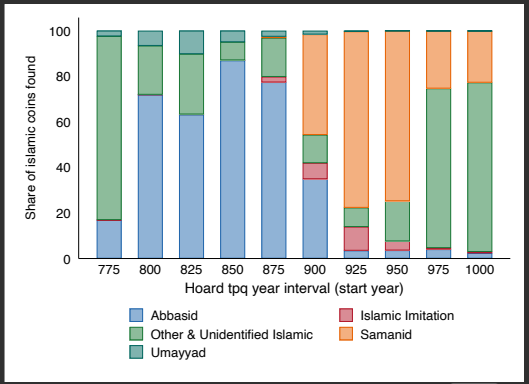
Back

Islamic coins by dynasty

By mint year

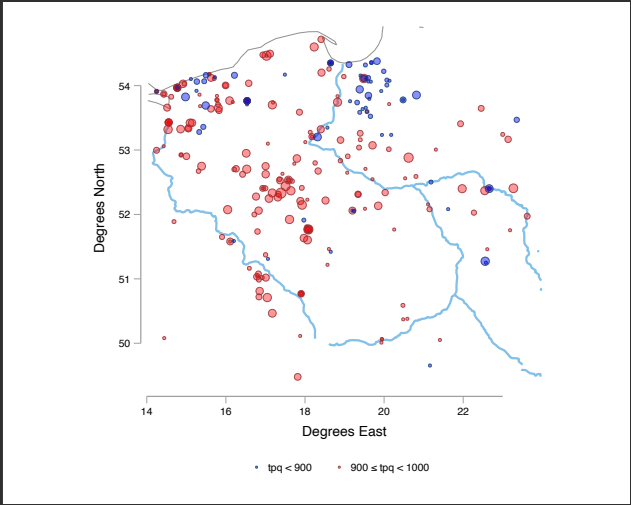


By hoard *tpq*



Back

Coin hoards in Poland, 9th–10th century



Trade disruption IV: controlling for expected exposure

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_{h}$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0318**				
	(0.0097)				
log Number of Coins		0.0905*			
		(0.035)			
Fraction of coins $<$ 20 y.o.			3.414*		
			(1.43)		
Fraction of coins $<$ 40 y.o.				3.223**	
				(1.23)	
Log coin age at deposit					-0.472**
					(0.12)
Hoard Size Controls			Yes	Yes	
$\mathbb{E}_1[\text{MinTravelTime}]_{it}$	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.881	8.764	5.660	6.833	12.74
R^2	0.163	-1.330	-16.43	-13.37	-11.68
Observations	429	429	911	911	105597

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Conflict timing fixed; disruption location varies in the assumed shock process.

Back

Trade disruption IV: alternative expected-exposure control

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_{h}$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0324**				
	(0.011)				
log Number of Coins		0.0999*			
		(0.042)			
Fraction of coins $<$ 20 y.o.			2.831*		
			(1.26)		
Fraction of coins $<$ 40 y.o.				2.853*	
				(1.23)	
Log coin age at deposit					-0.316**
					(0.098)
Hoard Size Controls			Yes	Yes	
$\mathbb{E}_2[\text{MinTravelTime}]_{it}$	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region × Time	Region × Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.955	5.986	5.068	5.385	9.697
R^2	0.143	-1.716	-11.14	-10.35	-4.964
Observations	429	429	911	911	105597

$^+$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Disrupted routes fixed; disruption assigned to the 9th or 10th century with equal probability.

Back

Trade disruption IV: including Rus territories

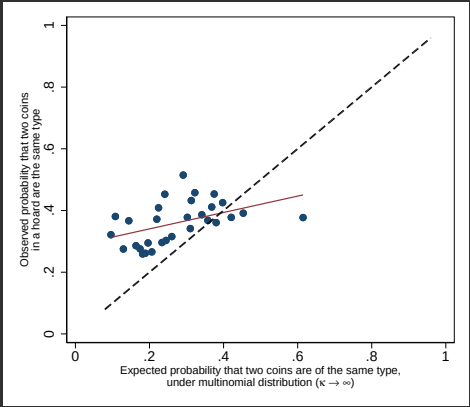
	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_h$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0775$^+$				
	(0.045)				
log Number of Coins		0.240*			
		(0.11)			
Fraction of coins $<$ 20 y.o.			2.825*		
			(1.11)		
Fraction of coins $<$ 40 y.o.				2.754**	
				(1.01)	
Log coin age at deposit					-0.561**
					(0.13)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.937	6.085	6.802	7.703	17.50
R^2	-1.729	-4.196	-10.40	-8.928	-11.91
Observations	429	429	911	911	105597

$^+$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Includes Rus territories. SEs clustered by region in (1)–(2), by coin group in (5).

Back

Overdispersion in coin types



- Vertical: probability that two coins from the same hoard have the same type
- Horizontal: independent-draw benchmark, excluding the focal hoard
- Above 45-degree line $\rightarrow$ excess within-hoard similarity

$\hat{\kappa} = 4.99$; independent multinomial limit: $\kappa \rightarrow \infty$.

Back

Polish river traffic estimates

Region	920–940	940–960	960–980	980–1000	Mean 940–980
Lower Vistula	0.77	0.77	0.73	1.05	0.85
Middle Vistula	0.15	0.24	0.10	0.20	0.18
Upper Vistula and Oder	0.40	0.61	0.55	0.74	0.63
Sum all three river regions	1.32	1.62	1.38	2.00	1.67

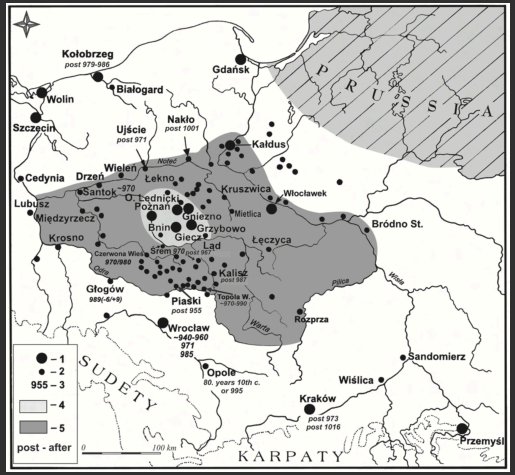
Back

Polish expenditure estimates

Region	920–940	940–960	960–980	980–1000	Mean 940–980
Lower Vistula	0.06	0.08	0.16	0.10	0.11
Middle Vistula	0.01	0.01	0.01	0.00	0.00
Upper Vistula and Oder	0.17	0.16	0.31	0.16	0.21
Sum all three river regions	0.24	0.25	0.48	0.26	0.33

Back

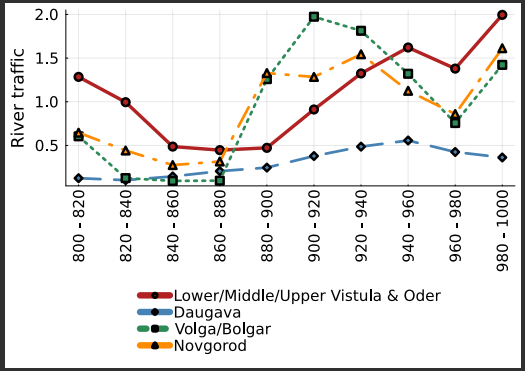
First Polish state under the Piast dynasty: Kara (2020)



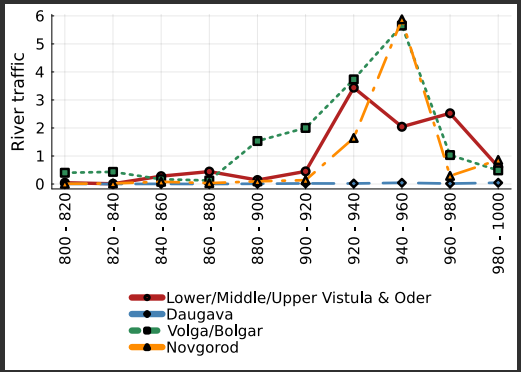
Poland, ca. 960. Source: Kara (2020).

Alternative trade model: within-hoard identification

Baseline



Within-hoard shares + gravity

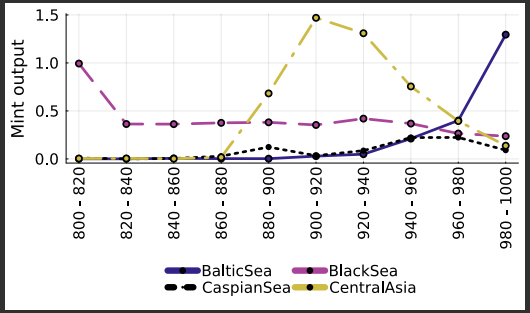


Alternative avoids identifying coin stocks from hoard counts; reconstructed Polish traffic is qualitatively similar.

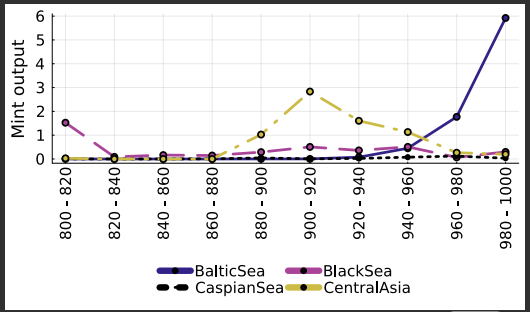
Back

Alternative trade model: estimated mint outputs

Baseline



Within-hoard shares + gravity



Back

Full parameter estimates

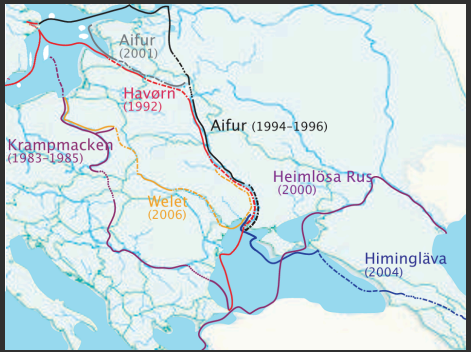
	Meaning	Estimate	95% CI	Interpretation
σ	Spatial decay of production tax value	0.071	[0.063, 0.073]	Break-even $\approx 1/\sigma \approx 14$ days
ω	Viking cost at the coastline	3.79	[2.27, 13.41]	$\approx 3.8\times$ a cell's production at the shore
κ	common Viking and Prague penetration scale	3.28	[2.40, 4.00]	both costs halve every ≈ 2.3 round-trip travel days
τ_r	toll per unit of (normalized) traffic	6.09	[5.39, 10.27]	a high-traffic river cell worth ≈ 7.1 cells
λ	frontier defense per edge	1.93	[1.69, 2.68]	one open frontier edge costs ≈ 1.9 cells
β_R	held-river defensive-side discount	0.94	[0.63, 1.07]	an eligible side of a held river cell costs $\approx 39\%$ of an open one
ρ	rival-polity cost at Prague	393.05	[128.22, 5000.00]	≈ 393.0 cell-equivalents at Prague

Back

Experimental archaeology: travel speeds



Krampmacken expedition on the Vistula, 1983.



Source: Edberg (2009).

Calibrated travel speeds: 8 km/h downstream; 1 km/h upstream and at portages.

Back

Definition of “territorial state”

Territorial state:

> *A centralized political organization that maintains durable control over a geographic area and its population, with the capacity to collect regular revenues and enforce its authority.*

Back

Coin flow model: Summary

Objective: estimate coins "traffic" on rivers. Strategy:

1. Law of motion for total coins in each location:

$$S_i(t+1) = (1-\lambda)S_i(t) + M_i(t) + Y_i(t) - X_i(t).$$

   S: coin stock; M: minting; $Y = wL$: income; X: expenditure.
   With no saving, $X_i(t) = (1-\lambda)S_i(t) + M_i(t)$, so $S_i(t+1) = Y_i(t)$.
2. Three assumptions:
   - 2.1 Coins are fungible ($\rightarrow$ mint date and mint don't matter).
     Then: *flows* of coins of each vintage are proportional to *stocks* of each vintage $\rightarrow$ can be identified from *distribution* of coin types in h
   - 2.2 Hoards are random samples from the population of coin vintages in each i, t
   - 2.3 $\mathbb{E}(\text{Q'ty of hoards}_{i,t}) \propto S_i(t)$

   Then: bilateral coin flows are identified from the data.
3. Assign each bilateral flow to the shortest river path between region centers.

Back

References i

Adamczyk, Dariusz. 2014. Silber und Macht: Fernhandel, Tribute und die piastische Herrschaftsbildung in nordosteuropäischer Perspektive (800–1100). Deutsches Historisches Institut Warschau, Wiesbaden:Harrassowitz.

Adamczyk, Dariusz. 2020. Monetarisierungsmomente, Kommerzialisierungszonen oder fiskalische Währungslandschaften?: Edelmetalle, Silberverteilungsnetzwerke und Gesellschaften in Ostmitteleuropa (800-1200). Harrassowitz Verlag.

Alesina, Alberto, and Enrico Spolaore. 1997. "On the number and size of nations." The Quarterly Journal of Economics, 112(4): 1027–1056.

Allen, Robert C., Mattia C. Bertazzini, and Leander Heldring. 2023. "The Economic Origins of Government." American Economic Review, 113(10): 2507–2545.

Allen, Treb. 2023. "The topography of nations." National Bureau of Economic Research.

Biermann, Felix. 2021. "Archaeological Evidence for Slavery Among the Early Medieval North-Western Slavs." In The Archaeology of Slavery in Early Medieval Northern Europe: The Invisible Commodity. 141–160. Springer.

Boehm, Johannes, and Thomas Chaney. 2025. "Trade and the End of Antiquity."

Borusyak, Kirill, and Peter Hull. 2023. "Nonrandom exposure to exogenous shocks." Econometrica, 91(6): 2155–2185.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2022. "Quasi-experimental shift-share research designs." The Review of economic studies, 89(1): 181–213.

References ii

Brather, Sebastian. 2007. “Counted and Weighed Silver: The Fragmentation of Coins in Early Medieval East Central Europe.” In Post-Roman Towns, Trade and Settlement in Europe and Byzantium. Vol. 1, 451–471.

Channon, John, and Robert Hudson. 1995. The Penguin Historical Atlas of Russia. London:Penguin Books.

Dal Bó, Ernesto, Karolina Hutková, Lukas Leucht, and Noam Yuchtman. 2025. “Dissecting the Sinews of Power: International Trade and the Rise of Britain’s Fiscal-Military State, 1689–1823.” The Journal of Economic History, 85(2): 336–369.

Edberg, Rune. 2009. “Experimental „Viking Voyages” on Eastern European Rivers 1983–2006.” Situne Dei, 35–46.

Fernández-Villaverde, Jesús, Mark Koyama, Youhong Lin, and Tuan-Hwee Sng. 2023. “The Fractured-Land Hypothesis.” Quarterly Journal of Economics, 138(2): 1173–1231.

Flückiger, Matthias, Mario Larch, Markus Ludwig, and Luigi Pascali. 2024. “The Dawn of Civilization: Metal Trade and the Rise of Hierarchy.” CESifo CESifo Working Paper 10929, Munich, Germany.

Heldring, Leander. 2025. “The Origins of Government and the State.” Annual Review of Economics, 17: 77–97.

Izdebski, Adam, Sambor Czerwiński, Marek Jankowiak, Marcin Danielewski, Sabina Fiołna, Raphael Gromig, Piotr Guzowski, Negar Haghipour, Irka Hajdas, Piotr Kołaczek, et al. 2025. “Unbalanced social–ecological acceleration led to state formation failure in early medieval Poland.” Proceedings of the National Academy of Sciences, 122(18): e2409056122.

Jankowiak, Marek. 2018. “Silver Fragmentation: Reinterpreting the Evidence of the Hoards.” In Silver, Butter, Cloth: Monetary and Social Economies in the Viking Age. Oxford University Press.

References iii

Jankowiak, Marek. 2020. “Dirham flows into northern and eastern Europe and the rhythms of the slave trade with the Islamic world.” In Viking-Age Trade. 105–131. Routledge.

Kara, Michał. 2020. “Polish historiography and archaeology on the mechanisms behind the formation of the Piasts’ regnum.” Przegląd Archeologiczny, 68: 121–135.

Kilger, Christoph. 2008. “Kaupang from Afar: Aspects of the Interpretation of Dirham Finds in Northern and Eastern Europe between the Late 8th and Early 10th Centuries.” In Means of Exchange: Dealing with Silver in the Viking Age. , ed. Dagfinn Skre, 199–252.

Kolmogorov, Vladimir, and Ramin Zabih. 2004. “What Energy Functions Can Be Minimized via Graph Cuts?” IEEE Transactions on Pattern Analysis and Machine Intelligence, 26(2): 147–159.

Kowalski, Tadeusz. 2022. Ibrahim ibn Yaqub’s Account of his travel to Slavic countries as transmitted by Al-Bakri: with contemporary commentaries.

Mayshar, Joram, Omer Moav, and Luigi Pascali. 2022. “The Origin of the State: Land Productivity or Appropriability?” Journal of Political Economy, 130(4): 1091–1144.

Noonan, Thomas S. 1992. “Fluctuations in Islamic trade with Eastern Europe during the Viking age.” Harvard Ukrainian Studies, 16(3/4): 237–259.

Roach, Andrew. 2020. “The People Trafficking Princes: Slaves, Silver and State Formation in Poland.” Slavonica, 25(2): 132–156.

References iv

Sánchez de la Sierra, Raúl. 2020. “On the Origins of the State: Stationary Bandits and Taxation in Eastern Congo.” Journal of Political Economy, 128(1): 32–74.

Tentiuc, Ion. 2020. “On the Viking enclaves and their relations with the inhabitants of the Carpathian—Dniester region between the 9th and the 11th centuries.” Dacia: Revue d’archéologie et d’histoire ancienne, LXII-LXIII: 250–286.

Tilly, Charles. 1992. Coercion, capital, and European states, AD 990–1992. Blackwell.

Леонтьев, А.Е., and Е.Н. Носов. 2012. “Восточноевропейские пути сообщения и торговые связи в конце VIII-X в.” In Русь в IX-X веках. , ed. Н.А. Макаров, 382–401. Институт археологии, РАН.